

Annex VIII — Perceived Real-Time Visual Information Delivery

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex. Version 2.0 (see §VIII.13).

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, all Presentation Surfaces, the Composition Engine, composition libraries in all provenance classes, the Session Composition Pool, the Prepared Edition, the Augmented Overlay, responsive HTML export, raster derivation, and all generation profiles defined by this Annex.

Dependencies: WR Graph™ structure, non-pre-satisfaction, routing-state exclusivity, and graceful degradation (Annex II); PoAC™ governance, evidence, mutation classes, and the Cloud Abstraction Boundary (Annex III); adapter and distribution governance (Annex IV); media reconstruction and converter governance (Annex V); WR Advance Preparation, WR Predictive Preparation, Real-Time Support, the Debrief, and retrieval privacy (Annex VI); and publisher artifact, manifest, grant, and Private Distribution Path machinery (Annex VII).

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VIII.0 Status, Scope, and Relationship to the Canon

This Annex specifies **Hybrid Composition**: the production of visually rich information artifacts — infographics in particular — by reasoning over information and composing reusable, annotated, quality-controlled elements, rather than synthesizing every artifact as one monolithic pixel image. It further specifies:

1. the Element Library and its governed optimization lifecycle;
2. the Composition Description and Composition Engine;
3. prepared, pooled, and fallback composition paths;
4. operator-selectable generation profiles;
5. responsive HTML as the primary structured Delivery Form;
6. deterministic raster and document derivations from the same canonical composition;
7. lifecycle-efficiency measurement from creation through consumption;
8. the Prepared Edition and Augmented Overlay.

The Annex distinguishes two layers that SHALL NOT be conflated:

- the **general composition architecture**, which may be implemented by any visual-generation or large-language-model system and may permit bounded generation of missing visual elements at request time; and
- the **WR governance profile**, in which privacy, provenance, reproducibility, and deterministic presentation impose stricter defaults while preserving informed operator choice.

This Annex does not redefine wire formats, capsule serialization, cryptographic primitives, transport, PoAE™ execution semantics, the WR Preparation lifecycle, or the media reconstruction model. It reuses those mechanisms and does not weaken them. Everything specified here is preparation, composition, delivery, or presentation. Nothing in this Annex creates, implies, expands, or pre-satisfies execution authority.

The core principle remains unchanged: **the system prepares, the human authorizes, and the system records cryptographic proof.**

VIII.1 Purpose, Motivation, and Architectural Principle

VIII.1.1 The Lifecycle Problem of Monolithic Infographics (Informative)

Information graphics are commonly produced and consumed as monolithic raster images. Whether created manually or synthesized by a generative image model, the final artifact fuses text, layout, icons, backgrounds, charts, and illustrations into one pixel surface. This form is convenient as a universal display object, but it carries avoidable costs across the complete lifecycle.

Generation cost. A complete image is synthesized even where most visual content consists of recurring and already-solved elements: containers, arrows, headings, color fields, icons, diagram primitives, typography, and standard illustrations. Full-image generation therefore spends model inference, accelerator time, memory bandwidth, and energy on recreating visual structures that could have been retrieved and reused.

Iteration cost. A correction to one value, label, icon, or section can require regeneration of the complete image. The cost of the smallest change approaches the cost of the whole artifact, and revisions may unintentionally alter regions that were already correct.

Integration cost. A website author receives a static end image rather than structured, responsive content. If the same visual structure is to become semantic HTML, accessible content, or a responsive component, substantial manual reconstruction may be required.

Transfer cost. A raster infographic may occupy hundreds of kilobytes or several megabytes. It is downloaded from the generation service, uploaded to an origin server, transferred through networks, and delivered to every uncached viewer. At large scale, small per-artifact inefficiencies accumulate across origin bandwidth, network transit, storage, caches, backups, and client retrieval.

Adaptation cost. A raster artifact is fixed to one aspect ratio and pixel geometry. Different device classes require scaling, cropping, or separate regeneration. Text cannot naturally reflow, and a mobile device may receive pixels intended for a desktop surface.

Semantic loss. Text fused into pixels is not natively selectable, searchable, machine-translatable, indexable, or consumable by assistive technology. Those properties can only be approximated through later reconstruction.

Caching loss. The same icon, diagram primitive, or background is repeatedly embedded inside unrelated images. The client cannot retrieve an element once and reuse it across artifacts because no element boundary remains.

The defect is therefore not limited to slow image generation. It is a lifecycle defect extending from reasoning and creation through revision, export, integration, transfer, rendering, reuse, accessibility, and later modification.

VIII.1.2 The Composition Principle (Informative)

This Annex takes the opposite construction:

A visual information artifact is composed from reusable, annotated, optimized elements. It is not required to be synthesized as one indivisible image.

A reasoning system determines the information architecture, selects suitable elements, arranges them within a constrained layout grammar, and writes the textual content. A retrieval and recommendation layer draws recurring visual substance from an Element Library. Only an element that the library cannot adequately supply requires new visual synthesis.

The approach separates four kinds of work:

1. **semantic reasoning** — what the artifact must communicate;
2. **composition reasoning** — which elements and layout best communicate it;
3. **asset provision** — retrieval of existing elements and bounded generation of genuinely missing elements;
4. **deterministic delivery** — responsive rendering and derivation of required output formats.

This separation permits model capacity otherwise spent on full-image pixel synthesis to be allocated to stronger reasoning, factual verification, responsive layout, accessibility, and quality control.

VIII.1.3 General Applicability Beyond WR (Informative)

The architecture is not limited to WR Desk™. A general-purpose LLM platform, image-generation provider, content-management system, or design tool may implement the same pattern:

1. generate a structured composition plan;
2. retrieve optimized components from a continuously improved library;
3. generate only missing components;
4. assemble a responsive HTML artifact;
5. derive images or documents from that same composition;
6. cache and reuse components across artifacts;
7. optimize the process through measured feedback loops.

In such systems, bounded real-time generation of missing elements may be a normal implementation choice. The WR-specific restrictions in this Annex are governance choices for a privacy- and determinism-oriented environment; they are not assertions that request-time generation is technically invalid in every system.

VIII.1.4 Scope and Delimitation (Normative)

The normative provisions bind the WR presentation chain: the preparation phases of Annex VI, the composition sources of VIII.6, the Delivery Forms of VIII.7, and the Presentation Surfaces of WR Desk™ and paired devices.

This Annex governs:

- how visual information artifacts are planned and represented;
- which visual elements may be used and under what provenance;
- how missing elements may be generated;
- how operator-selected generation profiles alter guarantees;
- how responsive HTML and raster outputs are derived;
- how efficiency and quality are measured;
- how visual artifacts are presented, audited, and degraded.

This Annex does not govern task execution, transport authorization, or the semantics of external actions. A composed artifact may present an action or recommendation, but it never grants authority to execute it.

VIII.1.5 The Hybrid Compute Split (Normative)

Composition work SHALL be placed according to its cost, privacy sensitivity, and reproducibility requirements.

- **Preparation phase:** broad semantic exploration, candidate generation, exemplar retrieval, library optimization, asset generation, responsive testing, and expensive quality evaluation.
- **Presentation phase:** deterministic branch evaluation, scored retrieval, bounded selection, slot binding, permitted refinement, rendering, and — only where an authorized generation profile permits it — bounded generation of missing visual elements.
- **External services:** only under the Cloud Abstraction Boundary of Annex III §7 and the retrieval-privacy rules of Annex VI §VI.7. Concrete WR Graph™ content, operational knowledge, and private composition-instance data SHALL NOT cross that boundary unless a separately defined and explicitly authorized disclosure path applies.

External reasoning is an accelerant, never a dependency. The complete composition chain SHALL remain operable in a local-only conforming mode.

VIII.1.6 The Default Inversion Rule (Normative)

Under the default **WR Deterministic Profile**:

Neural work that authors composition structure or visual assets SHALL occur in preparation, where its output is validated and captured as provenance-bound artifacts. The presentation

path SHALL be deterministic except for bounded selection, grammar-constrained textual generation, and the staged composition fallback defined in VIII.6.6.

The rule is an inversion of ordinary on-demand graphics generation. Information carries correctness obligations; presentation is therefore not the default place for unbounded creative synthesis.

The Inversion Rule is a default WR assurance rule, not a universal prohibition. It MAY be relaxed only through the explicit generation profiles of VIII.1.8 and VIII.5.4.

VIII.1.7 Composition, Not Whole-Image Synthesis (Normative)

Every displayed visual component SHALL originate from one of:

1. the host asset base;
2. an admitted Publisher Composition Library;
3. a Derived Composition Asset prepared locally;
4. a Live-Derived Composition Asset generated under an authorized profile;
5. operator-local data bound into typed slots.

A conforming implementation SHALL prefer retrieval and reuse over synthesis whenever an admitted element satisfies the declared need above the applicable quality threshold.

Full-image pixel synthesis MAY be offered as a non-primary fallback or export technique, but it SHALL NOT be treated as equivalent to Hybrid Composition unless the resulting artifact is reconstructed into a valid CD with resolvable element provenance and passes the same validation requirements.

Textual content MAY be generated at presentation time. Every factual value within generated text SHALL bind from provenance-bound sources: narrative may be generated; facts are bound.

VIII.1.8 Generation Profiles and Operator Choice (Normative)

The WR runtime SHALL expose at least the following profiles:

A. WR Deterministic Profile — default

- presentation-time visual synthesis disabled;
- only admitted, reconstructed, hash-bound assets displayed;
- missing decorative elements omitted, replaced by an admitted neutral element, or queued for preparation;
- exact reproduction of displayed artifacts required;
- no additional external interaction caused by rendering.

B. WR Local Live Generation Profile — optional

- bounded local generation of missing visual elements permitted;
- no composition data leaves the deployment;
- generated assets SHALL pass validation before display;
- generated assets carry `live-derived-local`: provenance;
- exact byte-for-byte regeneration is not guaranteed unless the complete generation state is retained, but the displayed output itself SHALL be hash-retained and reproducible as a rendered artifact;
- activation SHALL be scoped and revocable.

C. WR External Live Generation Profile — optional, highest disclosure class

- bounded use of an external generation service permitted;
- activation requires an informed operator decision under Annex III and the disclosure scope applicable to the request;
- the information dialog SHALL identify what data or abstractions may leave the deployment, the receiving class of service, expected cost, latency, provenance consequences, and reduced reproducibility;
- generated assets SHALL be reconstructed under Annex V before display and carry `live-derived-external`: provenance;
- activation SHALL be separately controllable from local live generation.

The operator MAY authorize a profile for one element, one artifact, one session, one context region, or a persistent policy scope. Persistent authorization SHALL remain discoverable and revocable. Viewing, waiting, or continuing a conversation SHALL NOT constitute authorization.

The system SHALL explain the guarantees changed by the selected profile in concrete terms: privacy, external disclosure, provenance, reproducibility, quality-floor status, latency, and cost. The dialog SHALL not use a generic warning as a substitute for an informed choice.

VIII.2 Definitions

Terms defined in Annexes II–VII apply unchanged. This Annex adds:

Composed Artifact. A visual information artifact produced from a validated Composition Description over hash-addressed elements and provenance-bound data. The canonical artifact is structured; one or more rendered Delivery Forms may be derived from it.

Composition Description (CD). The declarative, schema-constrained specification of a Composed Artifact: layout, element references, slots, responsive behavior, presentation policy, provenance, and output metadata. The CD is the unit of validation, staging, derivation, and audit.

Composition Engine (CE). The governed renderer transforming a valid CD into one or more Delivery Forms. Its identity includes version, content hash, rendering subset, constraint-solver version, and supported Render Profiles.

Composition Grammar. The permitted composition space: layout structures, slot types, responsive constraints, equivalence classes, transformations, and validation rules.

Element Library. The total set of reusable composition elements available to the CE across host, Acquired, Derived, and authorized Live-Derived provenance classes.

Publisher Composition Library (PCL). Publisher-produced assets, grammars, exemplars, domain presentation policy, optimization metadata, and media references published and admitted under Annex VII and PoAC™.

Derived Composition Asset. A locally prepared and validated reusable visual or structural element carrying `derived:` provenance.

Live-Derived Composition Asset. A missing element generated during an active request under an operator-authorized generation profile, validated before display, hash-retained, and carrying `live-derived-local:` or `live-derived-external:` provenance.

Element Annotation Record. Machine-readable metadata describing an element's semantics, style, geometry, responsive behavior, accessibility properties, provenance, quality metrics, licensing, and compatibility constraints.

Gap Detection. Deterministic or bounded-model identification that no admitted library element meets a declared need above the applicable fitness and quality thresholds.

Delivery Forms. Outputs derived from one canonical CD and its asset set. Responsive HTML is the primary form; raster snapshots and document forms are secondary derivations.

Responsive HTML Package. A passive, self-contained HTML/CSS representation plus its hash-addressed asset manifest, responsive rules, accessibility metadata, and integrity manifest.

Raster Snapshot. A pixel rendering of a CD produced by a governed Rasterizer under a declared Render Profile.

Render Profile. A versioned set of parameters governing a rendering derivation, including viewport, device pixel ratio, locale, color space, font set, background, output dimensions, format, encoder, compression, and quality settings.

Delivery Optimizer. The local deterministic component selecting an available Delivery Form for a target environment under quality, capability, privacy, transfer, and rendering-cost constraints.

Candidate Set. Event-bound, pre-validated CDs produced during a Preparation Run and selected by Branch Conditions.

Branch Condition. A deterministic predicate over locally available runtime data selecting a prepared candidate without inference.

Session Composition Pool. Session-scoped pre-generated and pre-validated CDs serving Real-Time Support through deterministic fitness matching.

Fitness Function. A deterministic, reproducible scoring algorithm ranking elements or composition candidates against a structured need descriptor.

Refinement Vocabulary. The closed set of deterministic transformations a grammar permits on an already validated candidate.

Prepared Edition. The editorial visual presentation form of the Morning Digest and Preparation Briefs.

Augmented Overlay. The operator-toggleable interaction layer rendered over passive composed artifacts.

Trigger provenance. `declared` for confirmed preparation and `inferred` for Preparation Queue hypotheses.

Quality Floor. The minimum acceptable correctness, legibility, accessibility, responsive stability, and visual-integrity thresholds below which an element or composition SHALL not be presented at full visual authority.

Lifecycle Efficiency Record. The measurement record comparing generation, revision, package size, transfer, cache reuse, rendering, and quality against a declared baseline.

VIII.3 The Element Library and Its Optimization Lifecycle

VIII.3.1 Element Classes and Annotation (Normative)

The Element Library MAY contain:

- layout containers and section structures;
- backgrounds, themes, and design tokens;
- icons, pictograms, and connectors;
- vector and raster illustrations;
- chart and diagram primitives;
- typographic resources;
- responsive templates and layout grammars;
- accessibility metadata and alternate textual descriptions;
- export and Render Profile compatibility declarations.

Every element SHALL be content-hash addressed and SHALL carry an Element Annotation Record containing at least:

1. semantic tags and domain association;
2. element class and declared use;
3. intrinsic geometry, aspect ratio, and permissible transformations;
4. supported responsive size classes and reflow behavior;
5. style family, color semantics, and theme compatibility;
6. accessibility properties, including alternate description where applicable;
7. provenance and admission state;
8. quality metrics and evaluation history;
9. licensing and export rights;
10. supported Delivery Forms;
11. usage, acceptance, rejection, and performance evidence.

Annotations inform scoring; they do not themselves grant authority.

VIII.3.2 Provenance Classes (Normative)

Library elements exist in four provenance classes:

1. **host** — local runtime assets;
2. **Acquired** — admitted PCL artifacts;
3. **Derived** — locally prepared reusable assets;
4. **Live-Derived** — request-time assets admitted for the current authorized scope and optionally proposed for later persistent admission.

All displayed elements SHALL be resolvable by hash. No provenance class carries elevated factual authority.

VIII.3.3 PCL Artifact Model, Distribution, and Admission (Normative)

A PCL is a set of publisher artifacts under Annex VII. Every artifact SHALL carry a signed manifest, be committed under the publisher's Manifest Root, and pass the Annex VII verification chain before staging.

Distribution follows the public repository or Private Distribution Path. Public-offering elements SHALL be publicly inspectable and auditable. Non-public elements remain hash-committed under the publisher's root and follow the evidentiary-reveal and attestation rules of Annex VII.

PCL admission is Class 3. Until admitted, its artifacts are unavailable to composition.

Media-class PCL assets SHALL traverse Annex V reconstruction. The CE composes only Reconstructed Artifacts. Text-pure structural and vector assets remain structured artifacts subject to schema and safety validation.

VIII.3.4 PCL Contents (Normative)

A PCL MAY contain:

- visual assets;
- layout grammars;
- responsive breakpoints and container rules;
- curated data-pattern-to-CD exemplars;
- domain presentation policy;
- media references;
- typography and localization policies;
- accessibility requirements;
- Render Profile recommendations;
- quality-evaluation fixtures;
- element-equivalence classes for deterministic refinement.

VIII.3.5 Constraining the Composition Space (Normative)

A publisher SHALL constrain composition through grammars wherever its PCL governs a domain. Free-form placement outside the applicable grammar is non-conformant.

Publishers SHOULD provide exemplars for recurring information patterns. Exemplar retrieval followed by slot binding and bounded adaptation SHOULD be preferred to unconstrained generation.

Where parametric composition competence is supplied through a LoRA-class adapter, Annex IV applies unchanged. Grammars and exemplars are the cheaper and more inspectable rungs and SHOULD be exhausted before adapter acquisition is proposed.

VIII.3.6 Domain Presentation Policy (Normative)

A domain presentation policy SHALL declare, where applicable:

- units, precision, and rounding;
- date, time, number, and locale conventions;
- reserved color meanings;

- source, timestamp, and confidence display requirements;
- chart-axis and scale rules;
- prohibited or misleading encodings;
- accessibility and contrast minima;
- mandatory context for values and forecasts;
- permitted transformations across Render Profiles.

The policy is applied deterministically at slot binding and validation. It never executes.

VIII.3.7 Media References (Normative)

Video and audio SHALL be integrated by attested reference and time range. Reconstruction and display follow Annex V. Retrieval SHALL comply with Annex VI privacy rules and SHOULD occur during the preparation window rather than in a correlatable presentation-time request.

VIII.3.8 Library Retrieval and Fitness Scoring (Normative)

Element retrieval MAY combine semantic embeddings, rules, Bayesian ranking, recommender systems, preference evidence, and learned quality estimates. The final admitted-element ranking used for deterministic presentation SHALL be reproducible from pinned inputs and scoring components.

The Fitness Function for elements SHOULD account for:

- semantic fit;
- style and domain compatibility;
- geometry and responsive fit;
- accessibility;
- quality score;
- provenance preference;
- expected package-size contribution;
- cache availability;
- licensing and export compatibility;
- historical operator feedback.

A highest-ranked element below the minimum threshold is a miss, not an acceptable match.

VIII.3.9 Gap Detection and Missing-Element Generation (Normative)

Gap Detection SHALL occur only after admitted retrieval has failed to satisfy the declared need above threshold.

A gap record SHALL identify:

1. the required semantic role;
2. geometry and responsive constraints;
3. applicable style and domain rules;
4. the reason retrieved candidates failed;
5. whether the element is informational, structural, or decorative;
6. the selected generation profile;
7. the quality and validation tests required before use.

Under the Deterministic Profile, the gap SHALL be resolved by omission, an admitted neutral substitute, degradation, or preparation queueing.

Under an authorized live-generation profile, only the missing element SHOULD be synthesized. Regeneration of the complete artifact is non-preferred unless the composition structure itself is invalid or no partial path exists.

VIII.3.10 Library Lifecycle and Optimization Loops (Normative)

The Element Library is a governed learning system, not a static asset folder.

The Night Pipeline MAY perform:

- coverage analysis over recurring composition gaps;
- element de-duplication;
- quality re-evaluation;
- annotation enrichment;
- responsive compatibility testing;
- cache-value estimation;
- format and compression optimization;
- proposal of new assets, grammars, exemplars, or equivalence classes;
- controlled benchmark experiments inspired by automated research loops.

Optimization SHALL be quality-constrained. No reduction in byte size, latency, or compute is acceptable if the applicable Quality Floor is violated.

Evidence accumulation is Class 1; deterministic re-ranking is Class 2; generation and staging of local Derived assets from admitted content is Class 2; acquisition of external assets is Class 3; persistent promotion of Live-Derived assets is governed by their source and disclosure path under VIII.5.4.

VIII.4 The Composition Description and Reasoning Outputs

VIII.4.1 The Three Reasoning Outputs (Normative)

Model reasoning contributes exactly three primary output classes:

1. **element selection** — references to elements or declared gaps;
2. **arrangement** — binding those elements into a grammar-permitted structure;
3. **textual content** — headings, labels, explanations, summaries, and narrative over provenance-bound facts.

Pixel-level synthesis is an asset-production activity, not a fourth composition-reasoning output.

VIII.4.2 CD General Requirements (Normative)

A CD SHALL:

- validate against the applicable Composition Grammar;
- be text-pure, referencing assets by identifier and hash;
- be self-contained with respect to its declared asset set;
- declare responsive behavior;
- declare Delivery Forms and supported Render Profiles;
- include provenance, confidence, and generation-profile metadata;
- contain no executable content;
- require no inference during deterministic rendering.

Model-produced CDs SHALL use schema-enforced decoding. A CD failing validation SHALL be discarded or explicitly regenerated; it SHALL NOT be silently repaired.

VIII.4.3 Slots and Fact Binding (Normative)

Templates SHALL declare typed slots. Slot binding is a pure operation: insert a provenance-bound value, format under domain policy, validate, and render.

Slots SHALL distinguish:

- factual values;

- generated narrative;
- citations and source lines;
- media elements;
- optional decorative elements;
- interactive affordance anchors rendered by the surrounding Presentation Surface.

Generated narrative MAY vary. Bound factual values SHALL not be invented by the narrative generator.

VIII.4.4 Responsive Constraints (Normative)

A CD SHALL define responsive behavior through a versioned set of constraints, including:

- grid and flow rules;
- min/max widths;
- breakpoints or container-query conditions;
- element priority and permissible omission;
- text expansion budgets;
- image-source selection;
- permitted stacking and reordering;
- typography scaling;
- fixed relationships that SHALL not be broken.

Responsive adaptation SHALL preserve information meaning and source association. A mobile layout MAY reorder presentation for readability but SHALL not change the logical or evidentiary relationship between elements.

VIII.4.5 CD Output Manifest (Normative)

Every CD SHALL carry or reference an output manifest containing:

- CD version and hash;
- grammar and policy versions;
- asset manifest;
- CE and Rasterizer identities;
- supported Render Profiles;
- default Delivery Form;
- default raster format;
- accessibility metadata;
- generation profile and authorization reference;
- export-rights status;
- derivation and integrity hashes.

VIII.5 Provenance, Generated Assets, and Informed Authorization

VIII.5.1 Trigger Provenance and Confidence (Normative)

Every CD SHALL declare:

- `trigger_provenance` $\in$ {declared, inferred};
- `confidence` or `declared status`;
- `preparation_ref` to the applicable Preparation Consent, Preparation Run, or Queue evidence chain.

These fields bind to visual salience under VIII.8.3.

VIII.5.2 Per-Element Provenance (Normative)

Every element SHALL declare one origin:

- host:<asset-id>@<hash>;
- pcl:<publisher-id>:<artifact-id>@<hash>;
- derived:<asset-id>@<hash>;
- live-derived-local:<asset-id>@<hash>;
- live-derived-external:<provider-class>:<asset-id>@<hash>;
- local:<binding>.

The rendered artifact SHALL preserve the ability to resolve the origin, admission record, generation record, and decision state of every element. Composition influence is never silent.

VIII.5.3 Derived Composition Assets Prepared Locally (Normative)

During preparation and idle periods, the deployment MAY generate reusable assets from admitted content.

Such assets SHALL:

1. be produced by a governed generation tool;
2. carry the tool identity, inputs, target context region, and content hash;
3. pass structural, visual, responsive, accessibility, and policy validation;
4. be staged as Class 2 with itemized Morning Digest reporting and rollback;
5. remain deactivatable without falsifying prior Derivation Records;
6. be regenerable from a retained recipe where technically possible.

Rejection deactivates future use and cascades to dependent staged compositions. It does not erase history.

VIII.5.4 Live-Derived Composition Assets (Normative)

A Live-Derived asset MAY be produced only under an active Local Live Generation or External Live Generation authorization.

Before activation, the system SHALL present an information dialog stating:

- whether generation is local or external;
- which information classes are used or disclosed;
- the receiving service class where external;
- the expected effect on determinism and reproducibility;
- the expected latency and cost class;
- the fact that the asset has not passed the same pre-admission history as a library element;
- the validation performed before display;
- the authorization scope and revocation method.

The authorization SHALL be recorded as a Consent Record. The generated asset SHALL not be displayed until it passes all required validation and Annex V reconstruction where applicable.

A Live-Derived asset is scoped by default to the current artifact. Persistent reuse requires a separate admission decision:

- locally generated from admitted content: MAY be proposed as a Derived asset, Class 2;
- externally generated or containing newly acquired external content: Class 3;
- generated under a scope prohibiting persistence: SHALL be discarded after audit-retention obligations are met.

VIII.5.5 Authority Parity (Normative)

Generated does not mean authoritative. Host, PCL, Derived, and Live-Derived elements are subject to the same factual binding, grammar, policy, and salience rules. A generated chart asserts no ground truth of its own; its values bind from declared sources.

VIII.5.6 Authorization Non-Coercion (Normative)

The Deterministic Profile SHALL remain fully functional. The runtime SHALL NOT degrade ordinary usability solely to pressure the operator into enabling live generation.

The operator SHALL be able to:

- decline without losing the textual answer;
 - choose local rather than external generation where available;
 - authorize only one missing element;
 - revoke persistent permission;
 - inspect the resulting provenance;
 - request deletion where compatible with audit and retention law.
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VIII.6 Prepared Composition Sources and Live Presentation

Prepared visual content reaches a Presentation Surface from three sources, in descending order of preparation specificity:

1. Candidate Sets;
2. the Session Composition Pool;
3. staged fallback composition, optionally augmented by authorized missing-element generation.

VIII.6.1 Candidate Sets and Branched Preparation (Normative)

For a Declared Event, the Preparation Run SHALL produce parameterized CDs covering plausible branches, each guarded by a deterministic Branch Condition.

Every candidate SHALL:

- validate at preparation time;
- declare pure-bind slots;
- inherit provenance and preparation lineage;
- include responsive behavior and supported Render Profiles;
- contain no unresolved informational gap required for factual correctness.

Candidate Set assembly from admitted content is Class 2. New external Sources or capabilities follow Class 3.

VIII.6.2 Presentation Resolution Path (Normative)

At presentation time, resolution proceeds:

1. evaluate Branch Conditions;
2. where no branch matches, rank Session Composition Pool candidates;
3. perform bounded selection if required;
4. bind current values;
5. apply permitted deterministic refinement;
6. detect remaining element gaps;
7. resolve gaps according to the active generation profile;
8. validate;
9. render the selected Delivery Form.

Implementations SHOULD target sub-second time-to-first-correct-display for prepared and pool paths. Telemetry SHALL record branch hit rate, pool hit rate, gap rate, selected profile, time-to-first-display, time-to-complete-composition, and Delivery Form.

VIII.6.3 Bounded Selection (Normative)

At presentation time, a local model MAY output only:

- a prepared-candidate identifier;
- a structured need descriptor;
- a binding map;
- a declared gap descriptor.

It SHALL NOT silently alter facts, provenance, or unrestricted composition structure.

VIII.6.4 Instantiation and Deterministic Refinement (Normative)

Instantiation is pure binding. Refinement is limited to the governing Refinement Vocabulary, including:

- equivalence-class element substitution;
- slot rebinding and policy formatting;
- omission of declared optional elements;
- reordering explicitly permitted by the responsive grammar;
- re-execution of the pinned constraint solver.

Every refined result SHALL revalidate before display.

VIII.6.5 Session Composition Pool (Normative)

The Session Composition Pool serves Real-Time Support where needs arise unpredictably within a known topic region.

VIII.6.5.1 Pool Generation

Pool candidates MAY be generated locally during session activation and idle windows under explicit compute, memory, and candidate-count budgets. They SHALL draw only on admitted content unless the active profile and consent permit otherwise.

VIII.6.5.2 Fitness-Scored Matching

The Fitness Function SHALL produce reproducible rankings from the need descriptor, candidate metadata, pinned scoring parameters, element annotations, and locally held preference evidence.

A match below threshold is a miss. The system SHALL not display a visually authoritative but semantically mediocre candidate merely because it is the best available.

VIII.6.5.3 Pool Lifecycle

Pool candidates are ephemeral. Usage and performance telemetry are Class 1 evidence. A recurring candidate pattern MAY be proposed as a persistent template, Class 3, under the recurring-pattern discipline of Annex III.

VIII.6.6 Prepared-Content Miss and Staged Fallback (Normative)

Where no prepared candidate satisfies the need, the system SHALL present a correct textual core immediately and proceed according to the active profile.

A. Deterministic Profile

1. show the textual core and available admitted elements;
2. compose a validated CD from available elements;
3. omit or neutrally substitute missing decorative elements;
4. degrade where an informational visual cannot be represented safely;
5. queue recurring gaps for preparation.

B. Local Live Generation Profile

1. show the textual core immediately;

2. generate only declared missing elements locally;
3. reserve their layout space where possible;
4. validate each generated element;
5. reveal it atomically when complete;
6. retain its output hash and provenance.

C. External Live Generation Profile

The Local Live Generation path applies, with additional disclosure, reconstruction, and consent requirements. No external result is displayed before reconstruction and validation.

Stage-1 text is content, not a placeholder. On timeout or validation failure, the composition SHALL remain complete in a simpler Delivery Form.

VIII.6.7 Perceived Real-Time Conformance (Normative)

A presentation is perceived-real-time conformant where the operator receives a correct and useful representation without an avoidable blank waiting interval. Completion of decorative or secondary elements MAY occur within the attention window, but factual content and action boundaries SHALL already be available.

VIII.7 Delivery Forms, Responsive Export, and Presentation Integrity

VIII.7.1 One Canonical Composition, Multiple Derivations (Normative)

The CD, its grammar, its policy set, and its hash-bound assets constitute the canonical composition. Delivery Forms are derivations of that composition; they are not independently authored artifacts.

A conforming system SHALL avoid separate semantic generation for HTML and image outputs. The same facts, textual content, element identities, and layout intent SHALL underlie every Delivery Form. Differences permitted by a Render Profile are responsive adaptation, pagination, scaling, compression, and declared omission — not semantic divergence.

Every derivation SHALL carry:

- the canonical CD hash;
- the asset-set root hash;
- CE or Rasterizer identity;
- Render Profile identifier;
- output hash;
- generation-profile provenance;
- creation time and locale;
- validation result.

VIII.7.2 Presentation Integrity Across All Forms (Normative)

The following rules apply to every Delivery Form:

1. **Validation before display.** No region is displayed before grammar, policy, provenance, and applicable accessibility checks pass.
2. **No silent factual revision.** Later correction appears as an explicit update, never as an unmarked replacement.
3. **Atomic element reveal.** A late element appears complete after validation; partially decoded or unvalidated imagery is not displayed.
4. **Reserved geometry.** Where late completion is permitted, the layout SHOULD reserve space to prevent disruptive layout shift.

5. **Source continuity.** A value and its source association SHALL remain intact across responsive transformations.
6. **Profile visibility.** Live-Derived content SHALL remain inspectably distinguishable in provenance even where it is visually integrated.
7. **Graceful degradation.** The chain is full responsive composition → reduced composition → table → text. A correct plain answer is complete; an indefinite progress indicator is not.

VIII.7.3 Responsive HTML Package — Primary Delivery Form (Normative)

Responsive HTML SHALL be the primary structured Delivery Form unless the Delivery Optimizer determines that the target cannot safely or efficiently consume it.

The package SHALL contain:

- passive semantic HTML;
- CSS within a pinned deterministic subset;
- a content-hash asset manifest;
- accessibility metadata;
- responsive rules and tested size classes;
- integrity metadata;
- no scripts, event handlers, forms, executable objects, trackers, or self-modifying content;
- no unapproved external resource references.

For the internal WR Presentation Surface, rendering SHALL cause zero network activity. Assets resolve exclusively from the local hash-addressed store.

For explicit website export, the package MAY be emitted as:

1. a self-contained bundle;
2. a content-addressed asset directory plus HTML/CSS;
3. an embeddable passive component package;
4. a server-renderable template representation.

The export manifest SHALL distinguish internal WR addresses from deployable web paths. Export SHALL not leak local paths, operator identifiers, graph identifiers, private source references, or internal provenance records not intended for publication.

VIII.7.4 Responsive Behavior and Device Classes (Normative)

Every Responsive HTML Package SHALL be validated against a declared device matrix. The minimum matrix SHOULD include narrow mobile, wide mobile, tablet, desktop, and large-display classes.

Validation SHALL cover:

- content overflow;
- text expansion and localization;
- reading order;
- contrast and font scaling;
- image resolution and cropping;
- source-line retention;
- chart legibility;
- touch-target spacing where surrounding interaction is present;
- layout stability;
- print or snapshot compatibility where declared.

A responsive version is not merely a scaled desktop image. Elements MAY stack, reflow, resize, or move within grammar-declared relationships.

VIII.7.5 Raster Snapshots — Secondary Delivery Form (Normative)

A Raster Snapshot SHALL be derived by a governed Rasterizer from the canonical responsive composition under a declared Render Profile.

The Rasterizer SHALL be tool-governed with pinned identity and content hash. The Render Profile SHALL specify at least:

- CSS viewport dimensions;
- output pixel dimensions;
- device pixel ratio;
- locale and directionality;
- font set and font hashes;
- color space;
- background and transparency;
- animation state, which SHALL be static;
- image-decoding policy;
- output format and encoder identity;
- compression and quality parameters.

The raster derivation SHALL occur only after the corresponding HTML composition validates.

VIII.7.6 Default and Optional Raster Formats (Normative)

The default raster snapshot format SHALL be **WebP** under a pinned encoder profile.

The default profile SHOULD use lossless or near-lossless settings for text-, icon-, and diagram-heavy artifacts. A domain policy MAY specify an alternative WebP profile for photo-dominant compositions.

A conforming implementation SHALL also support **PNG** as a lossless compatibility and archival export where required.

Implementations MAY support:

- AVIF for transfer-optimized web delivery;
- JPEG for opaque photo-dominant output;
- PDF for print and document workflows;
- SVG only where the complete composition is representable in the admitted safe vector subset.

Optional formats SHALL never replace the canonical CD and Responsive HTML Package in the audit chain.

VIII.7.7 Standard Render Profiles (Normative)

A deployment SHALL define versioned standard Render Profiles. The initial recommended set is:

- `mobile-narrow` — 360 CSS px reference width;
- `mobile-standard` — 390 CSS px reference width;
- `tablet-portrait` — 768 CSS px reference width;
- `desktop-standard` — 1440 CSS px reference width;
- `social-square` — 1080 × 1080 output;
- `social-portrait` — 1080 × 1350 output;
- `presentation-16x9` — 1920 × 1080 output;
- `print-a4` — print-profile rendering with declared margins and color policy.

These are named profiles, not hard-coded universal requirements. A deployment MAY define additional profiles, but every profile SHALL be versioned and reproducible.

VIII.7.8 Incremental Revision (Normative)

A revision SHALL operate at the smallest valid unit:

- text-slot update;
- data rebinding;
- element substitution;
- style-token update;
- section-level recomposition;
- complete recomposition only where dependencies require it.

The system SHALL compute a dependency impact set and SHALL not regenerate or retransmit unaffected assets merely because one region changed.

A changed CD SHALL receive a new hash. Unchanged element hashes remain stable, permitting cache reuse and precise visual diffs.

VIII.7.9 Cross-Artifact Caching and Transfer (Normative)

Hash-addressed reusable elements SHOULD be delivered with immutable caching semantics appropriate to the deployment environment. A consuming client may retrieve an element once and reuse it across every composition referencing the same hash.

The export package SHOULD support:

- deduplicated asset manifests;
- responsive source selection;
- compression appropriate to each media class;
- lazy loading for non-critical secondary assets where consistent with presentation integrity;
- explicit preloading of first-display assets;
- cache-independent fallback behavior.

Package optimization SHALL account for shared-cache state. The byte size of one HTML file alone is not a complete lifecycle measure.

VIII.7.10 Delivery Optimizer (Normative)

The Delivery Optimizer SHALL select a Delivery Form under a deterministic policy considering:

- target capability;
- responsiveness requirement;
- accessibility requirement;
- expected reuse and cache state;
- package and transfer size;
- client rendering budget;
- offline requirement;
- archival requirement;
- privacy and network policy;
- fidelity requirement;
- integration context.

Responsive HTML is preferred where reflow, semantics, accessibility, incremental revision, or reuse is valuable. A Raster Snapshot MAY be preferred where the target is static, constrained, archival, external, or where measured DOM and rendering cost exceeds the raster alternative.

The architecture SHALL not assume that HTML always consumes less client compute than a raster image. Efficiency is measured for the target process, not inferred from format labels.

VIII.7.11 Lifecycle Efficiency Measurement (Normative)

Claims of efficiency SHALL be supported by a Lifecycle Efficiency Record comparing the composed path with a declared baseline, normally an equivalent monolithic raster workflow.

Measurements SHOULD include:

- semantic-reasoning compute;
- visual-generation compute;
- time to first correct display;
- time to complete composition;
- bytes generated and stored;
- download and upload transfer volume;
- web-delivery transfer volume with cold and warm caches;
- client decode, layout, paint, and memory costs;
- revision cost for representative edits;
- cache reuse rate;
- responsive coverage;
- accessibility conformance;
- factual and visual quality scores.

An example such as a reduction from 1.5 MB to a few tens or hundreds of kilobytes MAY be documented informatively where measured. No universal factor-of-ten or energy-saving claim SHALL be presented as guaranteed without evidence for the declared workload.

VIII.7.12 Web Content Builder Use (Normative)

The Responsive HTML Package MAY be exported as a website content module. Such export SHALL preserve:

- semantic structure;
- responsive behavior;
- accessibility;
- source and attribution requirements;
- licensing constraints;
- integrity metadata;
- passive execution properties.

The export process SHALL generate a deployment report identifying:

- required assets;
- cache recommendations;
- CSS isolation strategy;
- supported browsers or rendering engines;
- any host-theme integration points;
- any features degraded for external compatibility.

This function converts the infographic generator into a structured visual content builder without changing the canonical composition model.

VIII.8 The Prepared Edition

VIII.8.1 Position (Normative)

The Prepared Edition is the visual presentation form of the Morning Digest and Preparation Briefs. It introduces no new governance object. Every item remains a Digest or Brief item with evidence, provenance, mutation class, and consent affordances.

The Edition changes presentation, not authority. It is assembled from validated CDs and SHOULD exhibit no operator-perceptible composition latency.

VIII.8.2 Priority-to-Salience Mapping (Normative)

Edition layout SHALL derive from local ranking through a deterministic mapping from ranking weight to layout weight, resolved against the governing grid grammar.

Reasoning determines what matters and what belongs together; the grid determines where and at what salience it appears.

VIII.8.3 Trigger Provenance and Visual Authority (Normative)

Declared and inferred content SHALL remain visually distinguishable.

- **Declared content** may occupy lead positions and the full salience range.
- **Inferred content** SHALL appear in a distinct advisory section with a salience ceiling below the declared range.

Confidence may order inferred items but SHALL not elevate them into declared visual authority. Viewing or expanding an inferred item does not accept it.

VIII.8.4 Multi-Publisher Pages (Normative)

Where multiple publishers appear on one Edition page:

1. ranking and salience are determined locally;
2. the host theme governs page-level coherence;
3. publisher styling is confined to the interior of its own elements;
4. per-element provenance remains discoverable;
5. no PCL field may influence relative prominence across publishers.

VIII.8.5 Deep-Dive Descent (Normative)

Every Edition item SHALL support inspection-depth expansion under Annex VI: full content, evidence, provenance, dependencies, and prepared supporting material without new retrieval.

Focused follow-up preparation remains a separately scoped Preparation Run. Deepening a proposal does not accept it.

VIII.8.6 Assembly Budget (Normative)

Inferred Edition content is bounded by the Digest budget and favors precision over recall. Declared preparation is owed completeness within its consented scope.

Full-fidelity composition is primarily warranted for declared content. Inferred content MAY be deliberately simpler, consistent with its advisory status and budget.

VIII.8.7 Delivery Forms for the Edition (Normative)

The Prepared Edition SHALL be available as responsive HTML within WR Desk™. It MAY additionally be exported under declared Render Profiles as WebP, PNG, PDF, or other permitted formats.

Every export remains hash-bound to the same Edition CDs and SHALL preserve the declared/inferred distinction.

VIII.9 The Augmented Overlay

VIII.9.1 Position and Toggle Discipline (Normative)

The Augmented Overlay is an interactive layer over passive composed artifacts. It provides annotations, navigation, automation suggestions, and spoken or pointer-based affordances.

It is operator-toggleable globally, per Presentation Surface, and per context region. No publisher, PCL, grant, or delivered artifact may activate it, prevent deactivation, or condition core functionality on its use.

VIII.9.2 Overlay Content (Normative)

Overlay content is composed and provenance-bound. Its passive visual layer is represented by CDs and rendered under the same HTML/CSS restrictions as ordinary artifacts. Interaction logic belongs to the trusted host runtime, not the exported artifact.

Publisher overlay content is Class 3 and region-bound at admission. Derived overlay annotations follow the Derived asset rules. New automation paths remain governed procedural capabilities.

VIII.9.3 Automation Suggestions and Execution Boundary (Normative)

An overlay suggestion pre-satisfies nothing. Every consequential action remains gated by PoAE™ at the point of effect.

Prepared actions retain their pinned template, schema validation, precise effect display, consent tap, and PoAE™ record. Overlay placement does not alter their authority.

VIII.9.4 Interaction Modalities and Evidence (Normative)

Speech SHALL be transcribed locally. Spoken confirmation is input, not an authorization factor unless a separate PoAE™ rule explicitly defines it.

Pointer, dismissal, expansion, and engagement telemetry are local evidence and SHALL not be observable by publishers. Ignored or dismissed content MAY dampen future surfacing.

The overlay MAY provide “prepare this” and “deepen this” affordances. These create evidence, declarations, or proposals through ordinary governance; they do not bypass preparation or training boundaries.

VIII.9.5 Overlay Export (Normative)

Website and raster exports SHALL exclude runtime overlay behavior unless the operator explicitly requests a static annotation layer. Static annotation export SHALL be flattened into the CD and lose all execution affordances.

VIII.10 Governance, Privacy, Auditability, and PoAC™ Classification

VIII.10.1 Grant Model and Execution Boundary (Normative)

All activity under this Annex is preparation, composition, delivery, or presentation. PCL delivery and updates ride on delivery/preparation rights. No CD, asset, Delivery Form, salience level, recommendation, or overlay item grants execution authority.

A prepared action remains executable only through the applicable human authorization and PoAE™ record.

VIII.10.2 Privacy and External Interaction (Normative)

All external interaction in service of composition is subject to Annex III §7 and Annex VI §VI.7.

Under the Deterministic Profile, presentation-time rendering causes no external interaction.

Under External Live Generation, the specific disclosure SHALL be authorized before transmission. The system SHALL minimize disclosure, prefer taxonomy-level or masked requests, and perform mandatory local instantiation where the returned result is generic.

A publisher SHALL not learn which library elements are used, when a composition is displayed, which Render Profile is selected, or how the operator interacts, except where the operator independently publishes or transmits an exported artifact.

VIII.10.3 Auditability and Reproduction (Normative)

For every displayed or exported artifact, the system SHALL retain or be able to resolve:

- the canonical CD;
- grammar and domain-policy versions;
- asset-set hashes;
- CE and Rasterizer identities;
- Render Profile;
- generation profile and authorization reference;
- preparation lineage;
- validation results;
- output hash;
- explicit updates or invalidations.

Under the Deterministic Profile, the artifact SHALL be exactly reproducible from retained inputs.

Under a live-generation profile, the exact generative process MAY be non-reproducible. In that case, the generated output actually displayed SHALL be retained by hash with sufficient provenance to reproduce the final artifact, and the audit record SHALL state that process-level regeneration is not guaranteed.

VIII.10.4 PoAC™ Assignment Table (Normative)

Operation	Class
CD rendering from validated inputs	1
Raster derivation under a pinned Render Profile	1
Responsive validation and static conformance tests	1
Session Pool generation, scoring, refinement, and discard from admitted content	1
Presentation and lifecycle telemetry recording	1
Evidence-based element re-ranking	2
Candidate Set and Prepared Edition assembly from admitted content	2
Local Derived asset generation and staging from admitted content	2
Deactivation and dependent cascade of a Derived asset	2
Local Live-Derived generation under an existing scoped authorization	2
Persistent promotion of a local Live-Derived asset from admitted content	2
PCL acquisition and admission	3
External Live Generation disclosure and interaction	3, unless covered by an explicit standing scope valid for the request
Persistent admission of externally generated content	3
New Source ingestion	3
Promotion of a recurring pool pattern to a procedural template	3
Domain-policy override	3
Admission of publisher overlay content or new automation templates	3

Nothing in this table permits new external knowledge or capability to enter the WR Graph™ without the Consent Record required by Annex III.

VIII.10.5 Retention and Deletion (Normative)

Audit retention SHALL be proportional to assurance requirements and applicable law. Deactivation SHALL be preferred to destructive deletion where prior artifacts depend on a retained hash.

Where an operator requests deletion of Live-Derived content, the system SHALL explain any conflict with legal, security, or audit retention and SHALL remove future-use references immediately where permitted.

VIII.11 Quality-Constrained Optimization and Evaluation

VIII.11.1 Optimization Objective (Normative)

Optimization SHALL seek better allocation of resources at equal or higher output quality. The optimization objective is multi-dimensional and SHALL not collapse into minimum file size or minimum latency alone.

The objective vector SHOULD include:

- factual correctness;
- semantic coverage;
- visual hierarchy;
- legibility;
- accessibility;
- responsive stability;
- operator comprehension;
- provenance completeness;
- generation latency;
- inference and rendering compute;
- package and transfer size;
- cache reuse;
- revision cost;
- integration effort.

Hard Quality Floor constraints dominate soft efficiency objectives.

VIII.11.2 Automated Research-Inspired Loops (Normative)

A deployment MAY run automated experimentation loops over admissible variants of:

- retrieval parameters;
- element ranking;
- template selection;
- responsive constraints;
- compression profiles;
- Render Profile parameters;
- cache and preload policy;
- layout grammar refinements.

Every experiment SHALL have:

1. a declared hypothesis;
2. a fixed evaluation set;
3. quality and safety constraints;
4. reproducible parameters;
5. measured results;
6. rollback;
7. governance classification;

8. a prohibition on autonomous expansion into new external Sources or capabilities.

The loop may discover better parameters. It does not grant itself new authority.

VIII.11.3 Accessibility and Localization (Normative)

Responsive HTML output SHALL support semantic reading order, selectable text, alternate descriptions, sufficient contrast, and keyboard-compatible surrounding controls.

Localization tests SHALL account for text expansion, right-to-left direction, locale-specific number and date formats, and font coverage. A layout valid only for the language in which it was generated is not generally responsive.

Raster exports SHOULD embed accessible descriptions in surrounding document metadata where the target format permits. The existence of a raster export does not waive the requirement to preserve an accessible structured form where available.

VIII.11.4 Security Evaluation (Normative)

Export and rendering tests SHALL include:

- script and active-content exclusion;
- external-reference detection;
- CSS subset conformance;
- font and media validation;
- path and identifier leakage checks;
- reconstruction enforcement;
- content-security-policy compatibility;
- hash-manifest verification;
- malformed-layout and resource-exhaustion tests.

VIII.11.5 Benchmark Baselines (Normative)

A claimed improvement SHALL name its baseline. Suitable baselines include:

- full-image generative synthesis;
- manually authored raster infographic;
- static raster web embedding;
- separately authored desktop and mobile artifacts;
- cold-cache and warm-cache delivery.

Quality equivalence SHALL be evaluated independently of efficiency. A smaller artifact that communicates less, obscures evidence, or fails accessibility is not an improvement.

VIII.11.6 Conformance Levels (Normative)

A deployment MAY declare:

- **Composition Conformance:** valid CD, Element Library, deterministic CE;
- **Responsive Delivery Conformance:** passive responsive HTML and device-matrix validation;
- **Multi-Form Conformance:** governed raster derivation and Render Profiles;
- **Lifecycle Optimization Conformance:** measurement, incremental revision, and cache-aware delivery;
- **WR Deterministic Conformance:** default profile, exact reproduction, zero render-time external interaction;
- **WR Informed Live Generation Conformance:** scoped operator choice, differentiated local/external modes, validation, provenance, and revocation.

A higher declaration SHALL include the requirements of lower levels on which it depends.

VIII.12 Relation to Prior Work, General Adoption, and Claimed Contribution (Informative)

Declarative UI systems, HTML/CSS rendering, design systems, component libraries, responsive web design, visualization grammars, content-addressed storage, image rasterization, recommender systems, constrained decoding, caching, and precomputation are established practice and are not individually claimed as novel.

The general architectural contribution expressed by this Annex is their coordinated application to generative visual information:

- treating the composition rather than the monolithic image as the canonical artifact;
- assigning model reasoning to information, selection, and arrangement;
- retrieving optimized visual elements from an annotated library;
- detecting gaps and confining generation to genuinely missing elements;
- continuously improving the library through measured, quality-constrained loops;
- deriving responsive HTML and multiple image or document forms from one composition;
- reducing revision, integration, transfer, and reuse costs across the complete lifecycle;
- permitting delivery optimization by target rather than assuming one format is universally best.

The WR-specific contribution is the governance layer applied to that architecture:

- attestation-bound library admission;
- per-element provenance;
- reconstruction of media-class assets;
- deterministic prepared-first presentation;
- a strict but non-paternalistic default;
- separate local and external live-generation profiles;
- informed, scoped, revocable operator choice;
- explicit handling of reduced reproducibility;
- trigger-provenance binding to visual authority;
- cross-publisher ranking exclusivity;
- exact audit of what was shown, from what, under which rules.

Large LLM and visual-generation providers may adopt the general architecture without adopting WR governance. In that environment, real-time generation of missing elements may be routine. The essential principle remains: **generate the irreducibly new part; retrieve and compose the rest; derive every required form from one structured source.**

VIII.13 Version History

Version	Date	Change
1.0	2026-07-11	Initial normative Hybrid Composition specification: preparation/presentation compute split; Inversion Rule; Composition Description; passive HTML/CSS rendering; PCL and Derived Composition Assets; Candidate Sets; Session Composition Pool; staged fallback; Prepared Edition; Augmented Overlay; PoAC™ classification; privacy and auditability.
1.1	2026-07-14	Terminology alignment with WR

Version	Date	Change
		Advance Preparation, WR Predictive Preparation, Real-Time Support, and Private Distribution Path. No intended semantic change.
2.0	2026-07-17	Full lifecycle revision. Adds the explicit problem statement for monolithic raster infographics; general applicability beyond WR; annotated and optimization-driven element retrieval; gap detection and generation limited to missing elements; distinction between the general architecture and WR assurance defaults; WR Deterministic, Local Live Generation, and External Live Generation profiles with informed, scoped, revocable operator choice; Live-Derived provenance and admission rules; responsive HTML as primary Delivery Form; deterministic WebP raster snapshots with PNG compatibility and versioned Render Profiles; website content-builder export; incremental revision; cross-artifact caching; Delivery Optimizer; lifecycle-efficiency measurement and evidence requirements; automated research-inspired quality-constrained optimization; accessibility, localization, and security evaluation; corrected PoAC™ classification and complete cross-reference restoration.

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FIGURE 1 / 4

THE MONOLITHIC RASTER PROBLEM

Today's infographic is typically delivered as one fused raster image:
layout, text, iconography, and decoration collapsed into pixels.

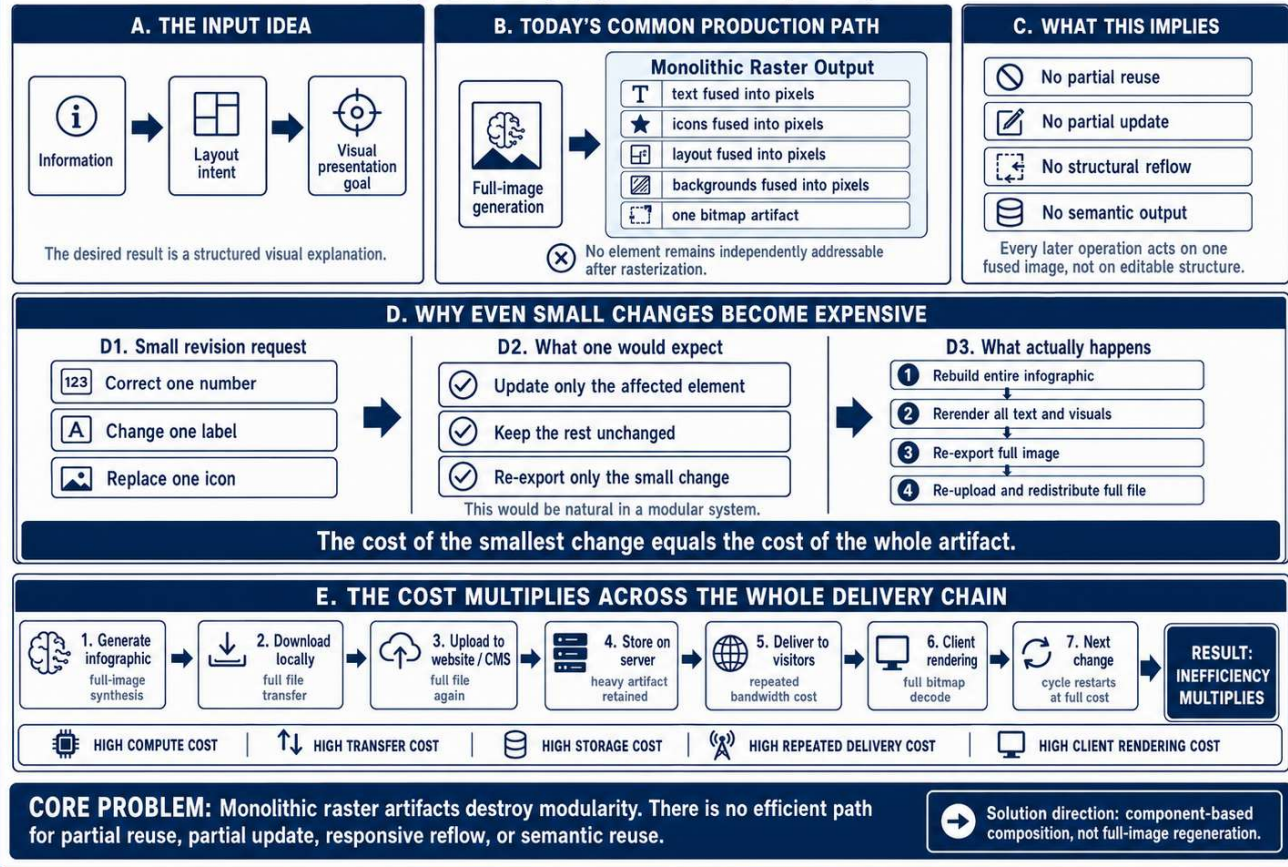


FIGURE 2 / 4

TINY CHANGE, FULL REGENERATION

In a monolithic raster pipeline, even a minor revision re-triggers the entire generation path.

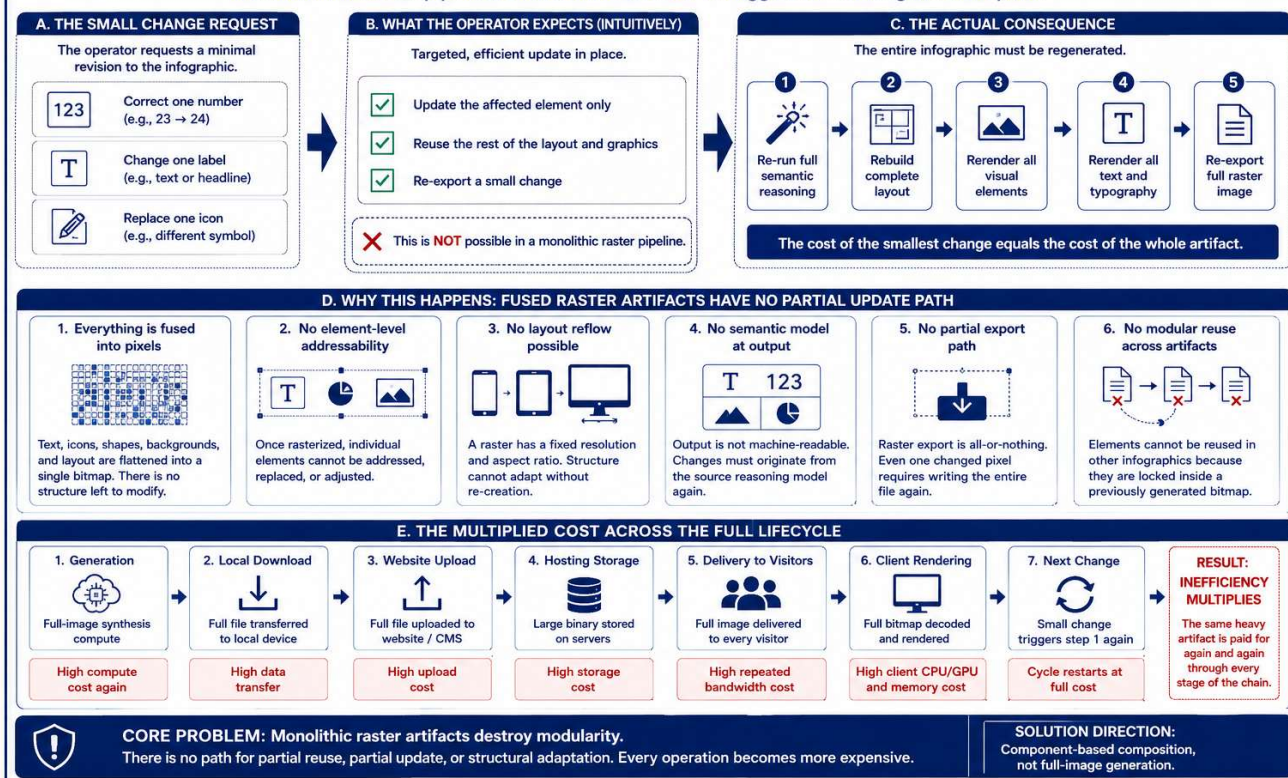


FIGURE 3 / 4

WASTE ACROSS THE FULL DELIVERY CHAIN

A large raster infographic is expensive not only to create, but also to move, host, and display.

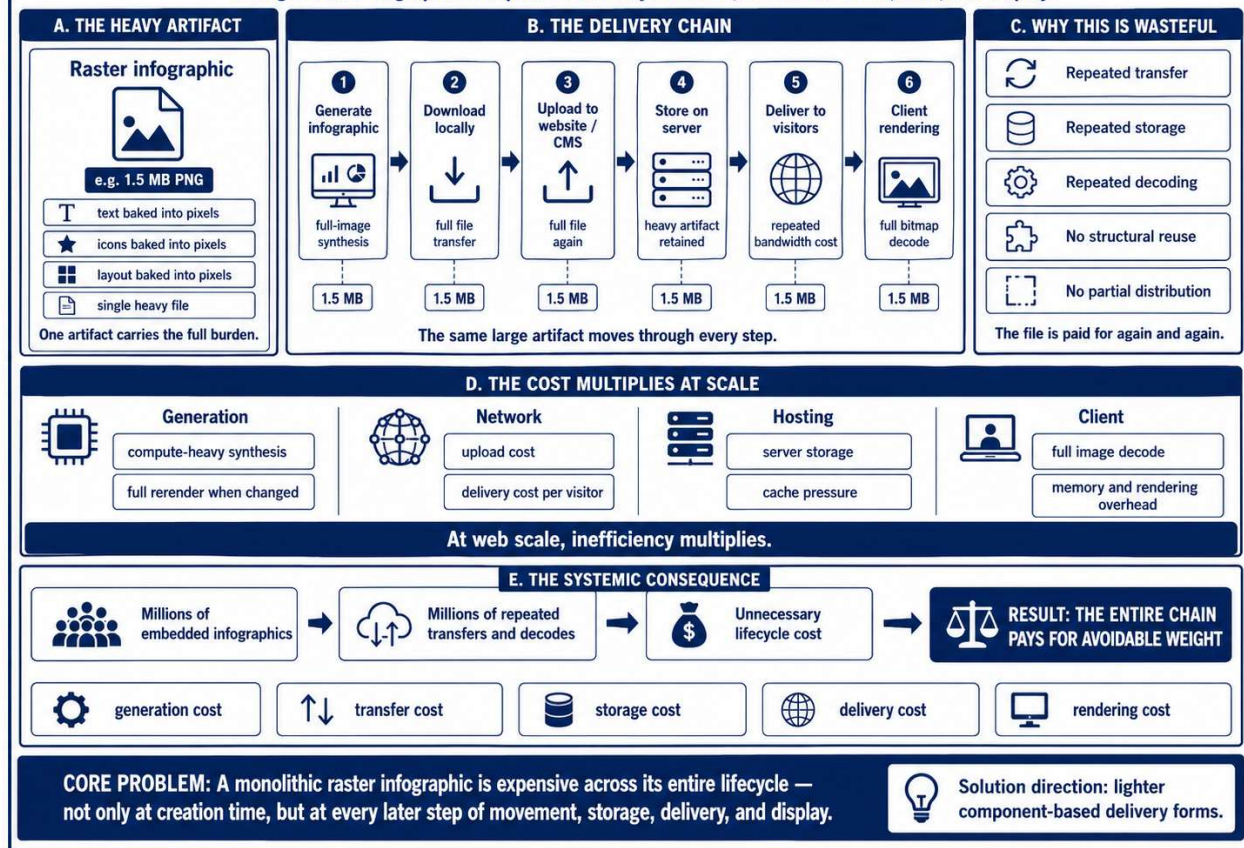


FIGURE 4 / 4

FIXED PIXELS, BROKEN RESPONSIVENESS, LOST SEMANTICS

A raster infographic is tied to one resolution and one fused pixel structure: it does not truly reflow, and its text no longer behaves like text.

